STUDENT PERFORMANCE MANAGER

ASSIGNMENT 3 - CODING DOCUMENTATION

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Register No.: 242BCA05

Subject: Software Engineering Concepts

Submitted to: Mr. Selwyn Paul J

Semester: 4

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GitHub Link: https://github.com/Gopikaa211

Website Link: https://student-performance-manager.vercel.app

Overview:

The Student Performance Manager is a web-based system developed using HTML, CSS, JavaScript, and Firebase. The application allows teachers to manage student records, mark attendance, and enter internal assessment marks. Students can log in to view their attendance and academic performance. Firebase Authentication is used for secure login, and Firestore is used for storing academic data.

System:

User (Teacher / Student)

↓

Frontend (HTML, CSS, JS)

↓

Firebase Authentication

↓

Firestore Database

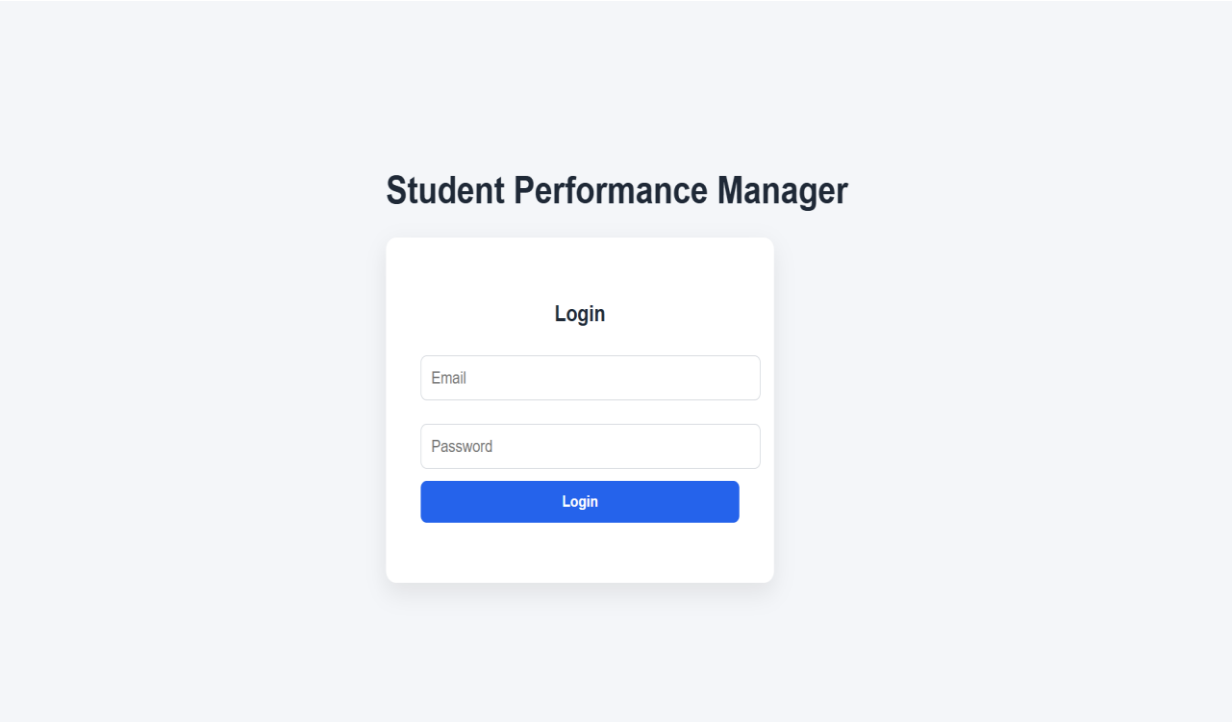
↓

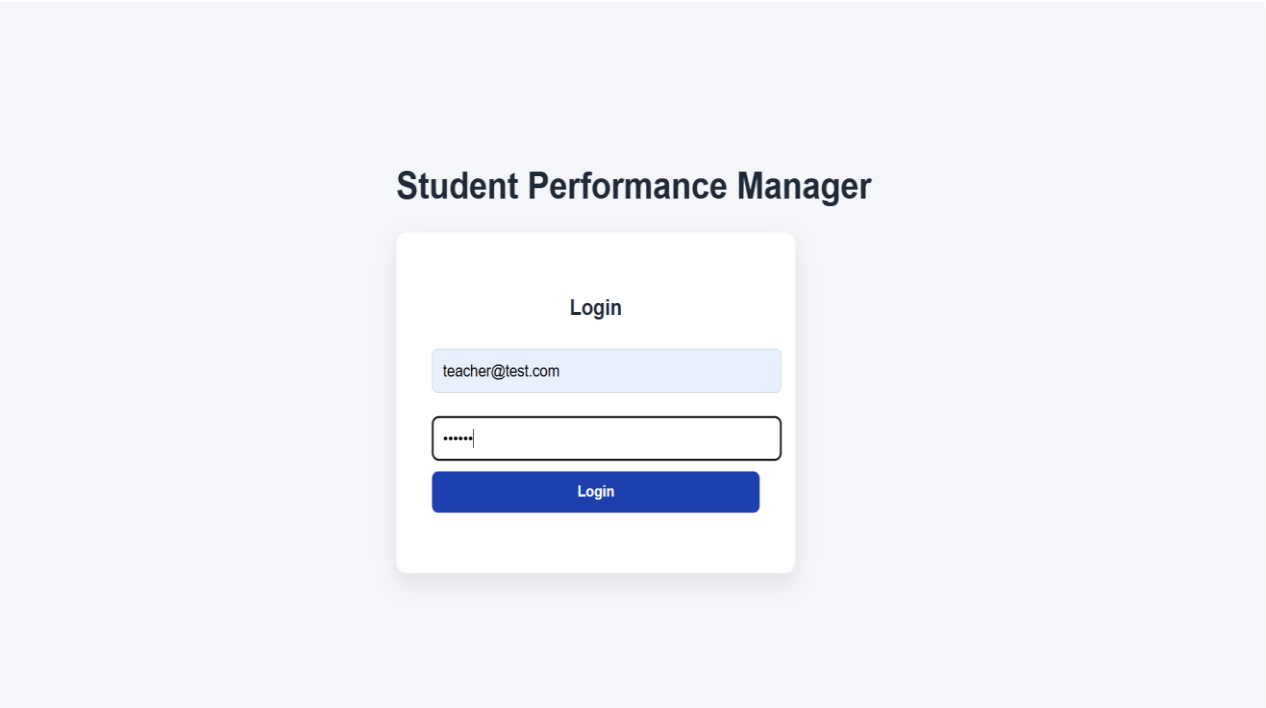
Student / Attendance / Results Data

The system follows a client–server architecture where the frontend interacts with Firebase services for authentication and data storage. Firebase provides authentication and a cloud database, allowing the application to securely manage users and store student academic records without maintaining a separate backend server.

Login Page

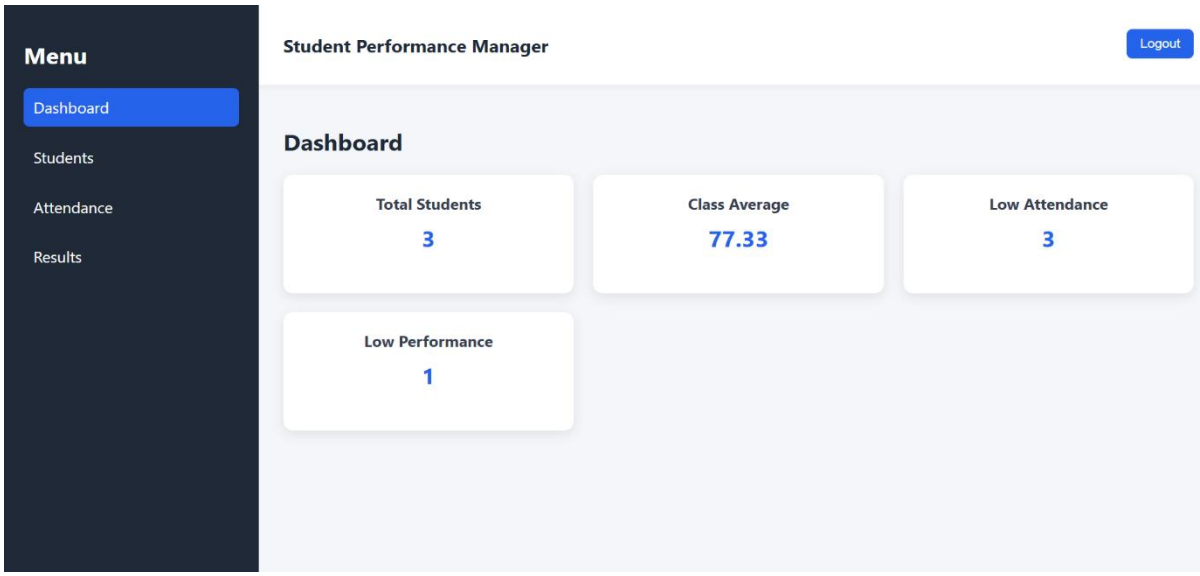
The login page allows teachers and students to access the system using their registered email and password. Firebase Authentication verifies the user credentials and redirects them to the appropriate dashboard.





Teacher Dashboard

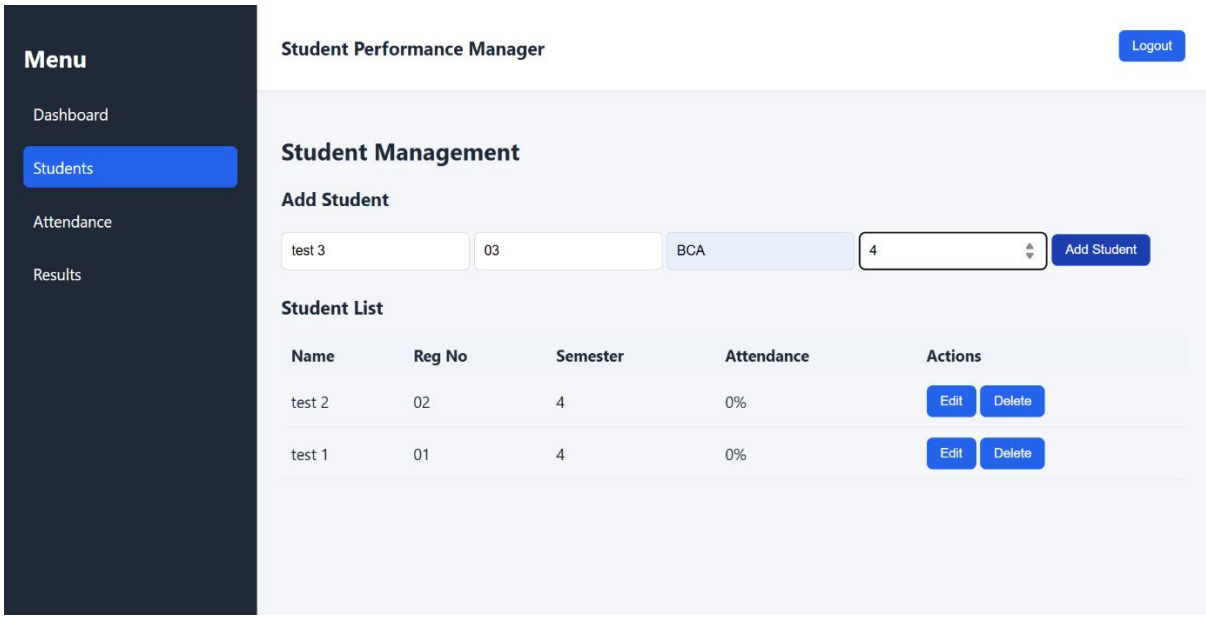
The teacher dashboard displays an overview of the academic data, including total students, class average, low attendance alerts, and low performance alerts.



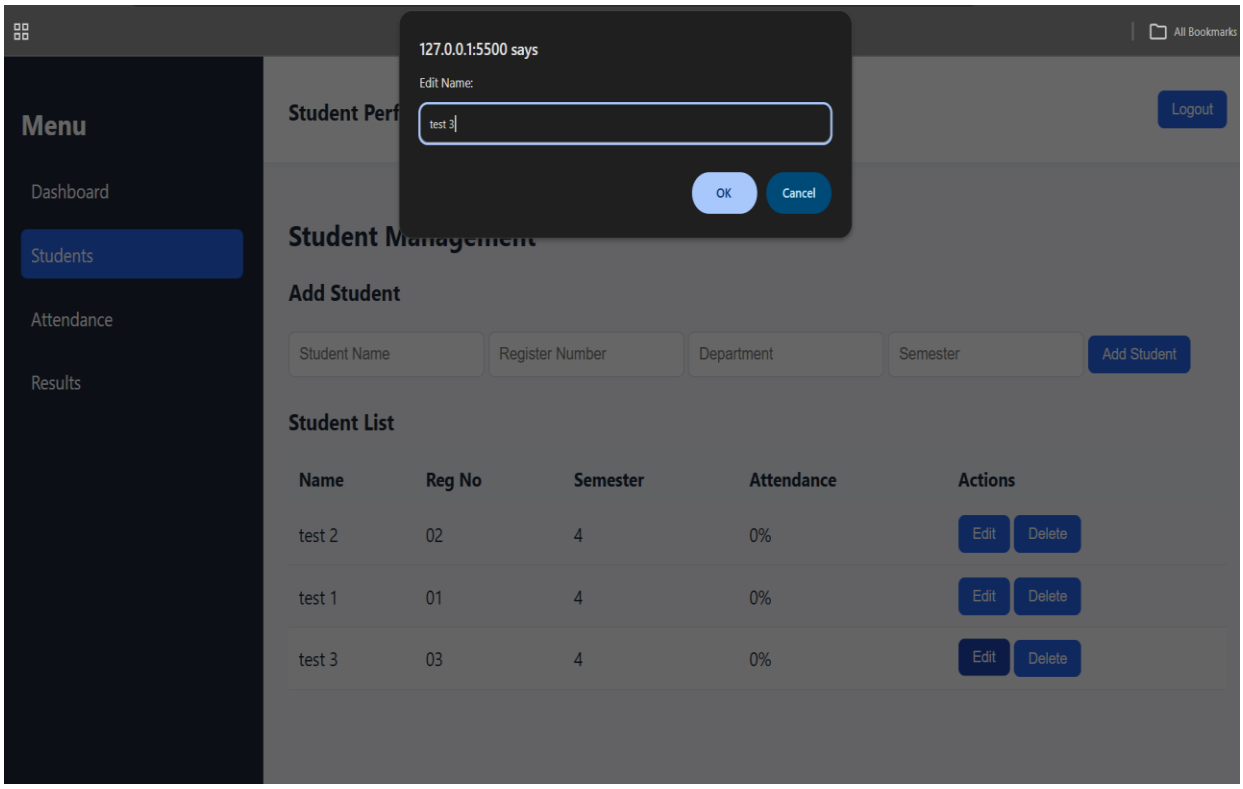
Student Management Page

The student management module allows teachers to add, update, and delete student records. These records are stored in the Firestore database.

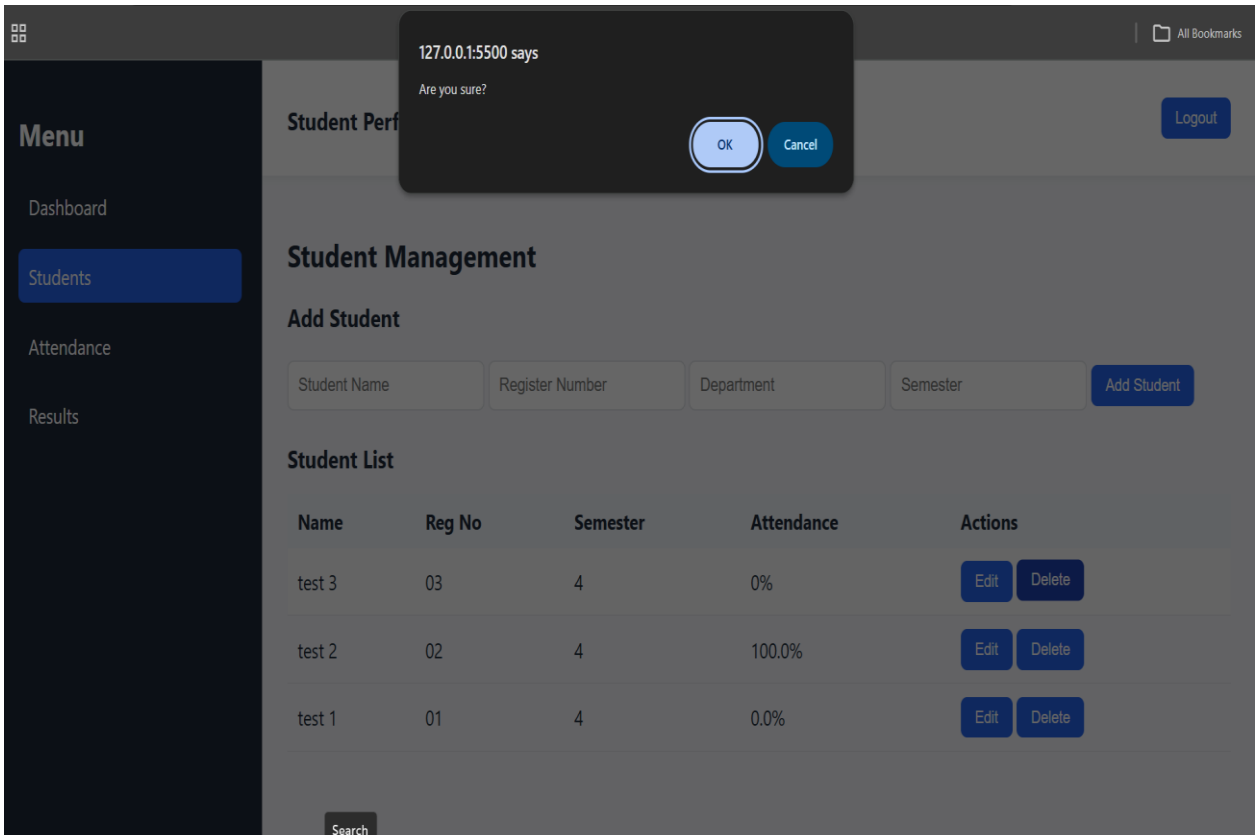
Adding a new student **test 3** to the student list



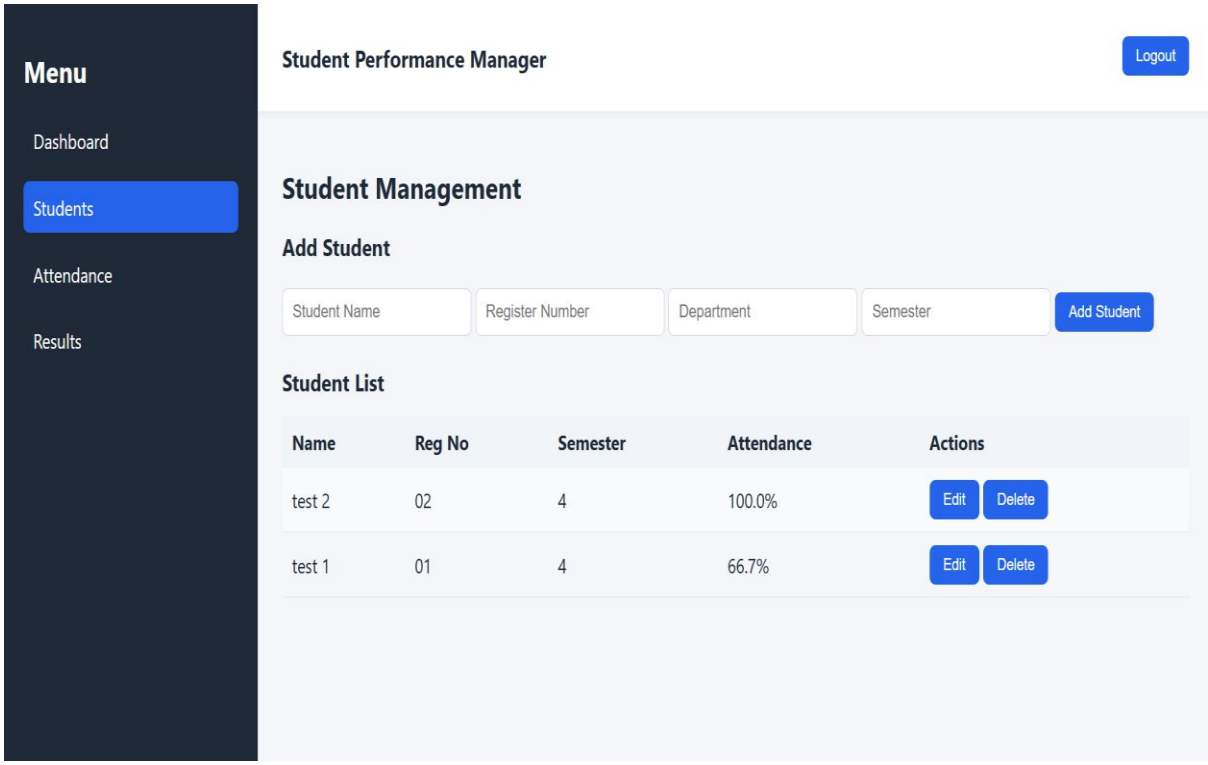
Editing the newly added student **test 3**



Deleting newly added student **test 3**

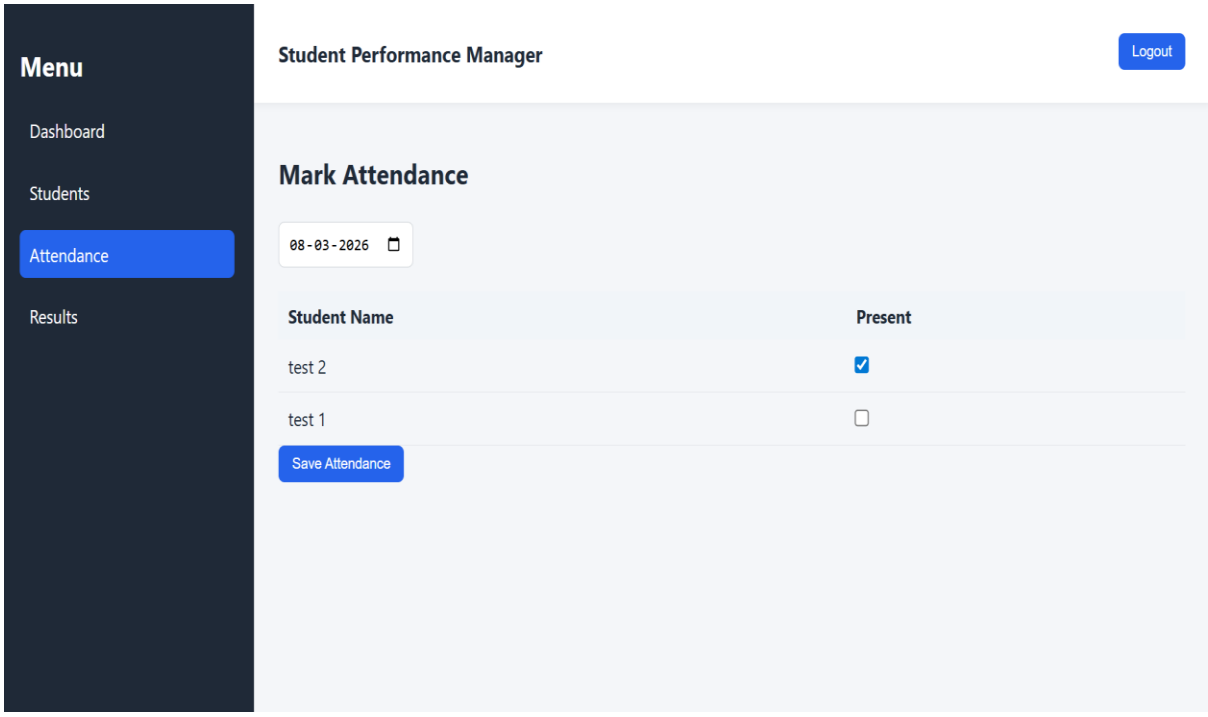


Student list after the CRUD operations

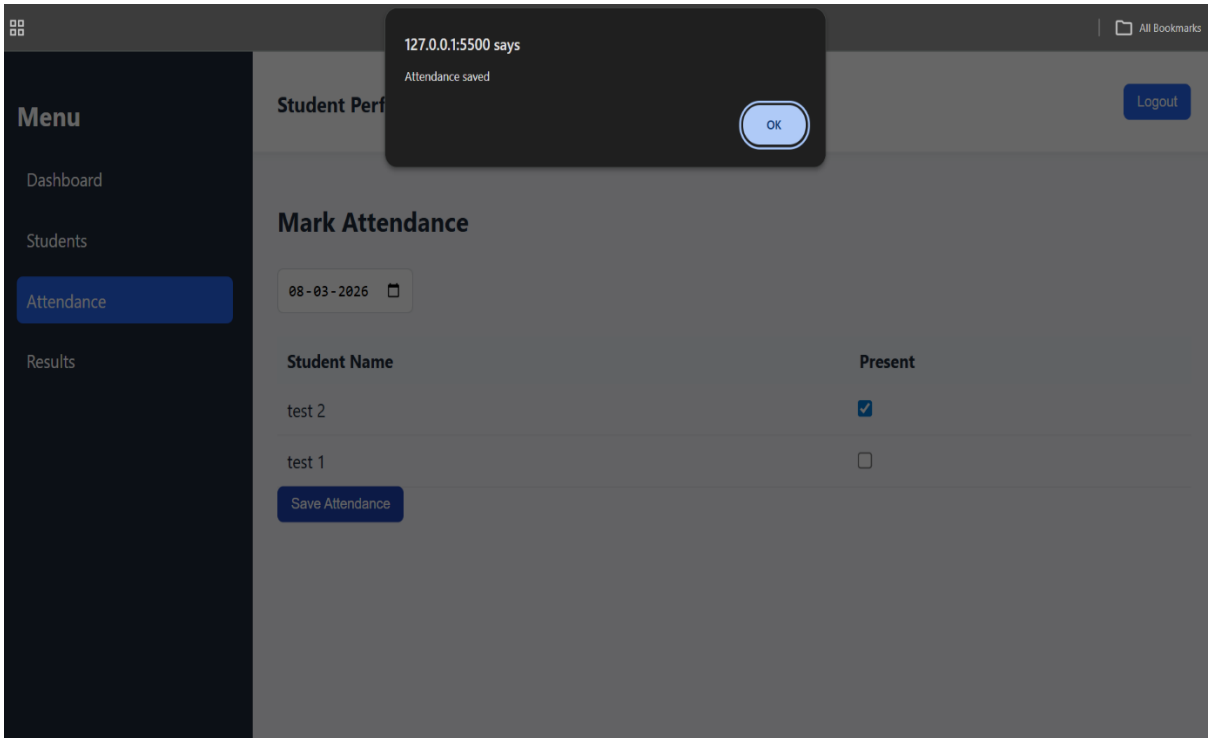


Attendance Page

The attendance module allows teachers to mark student attendance using checkboxes. Each attendance record is stored with the student ID and date.

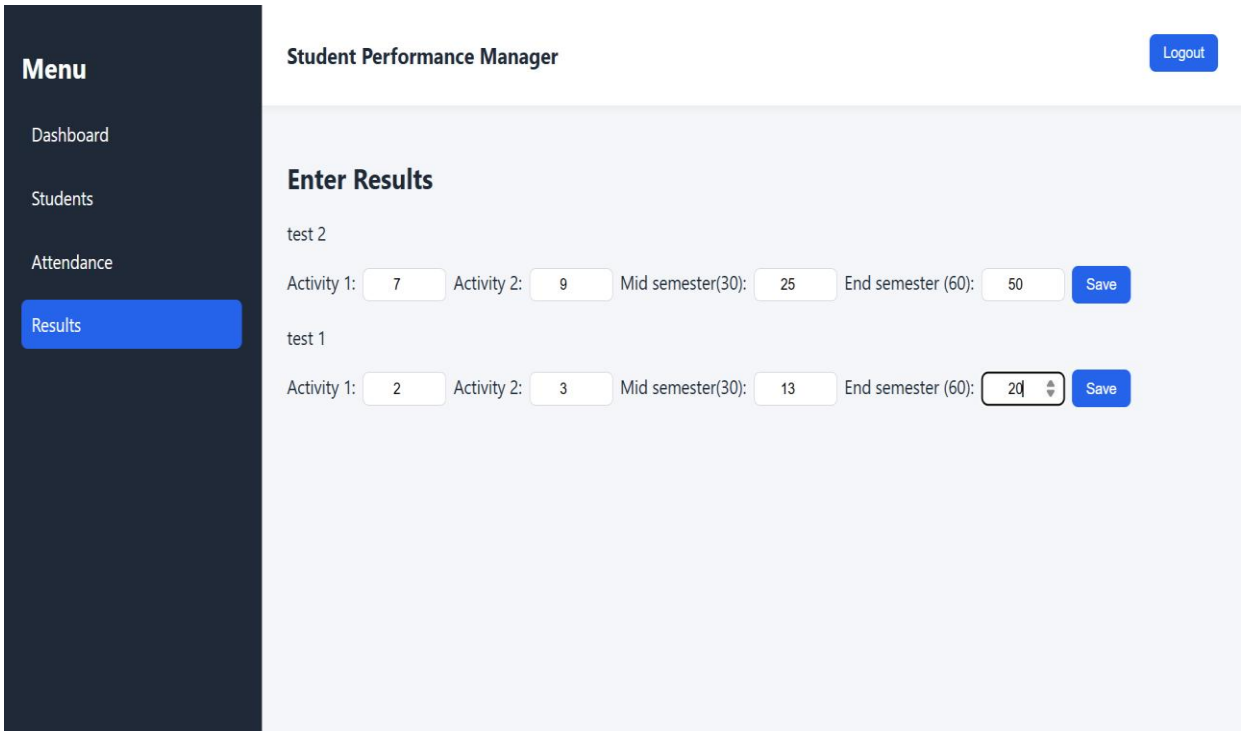


Attendance **saved** to database

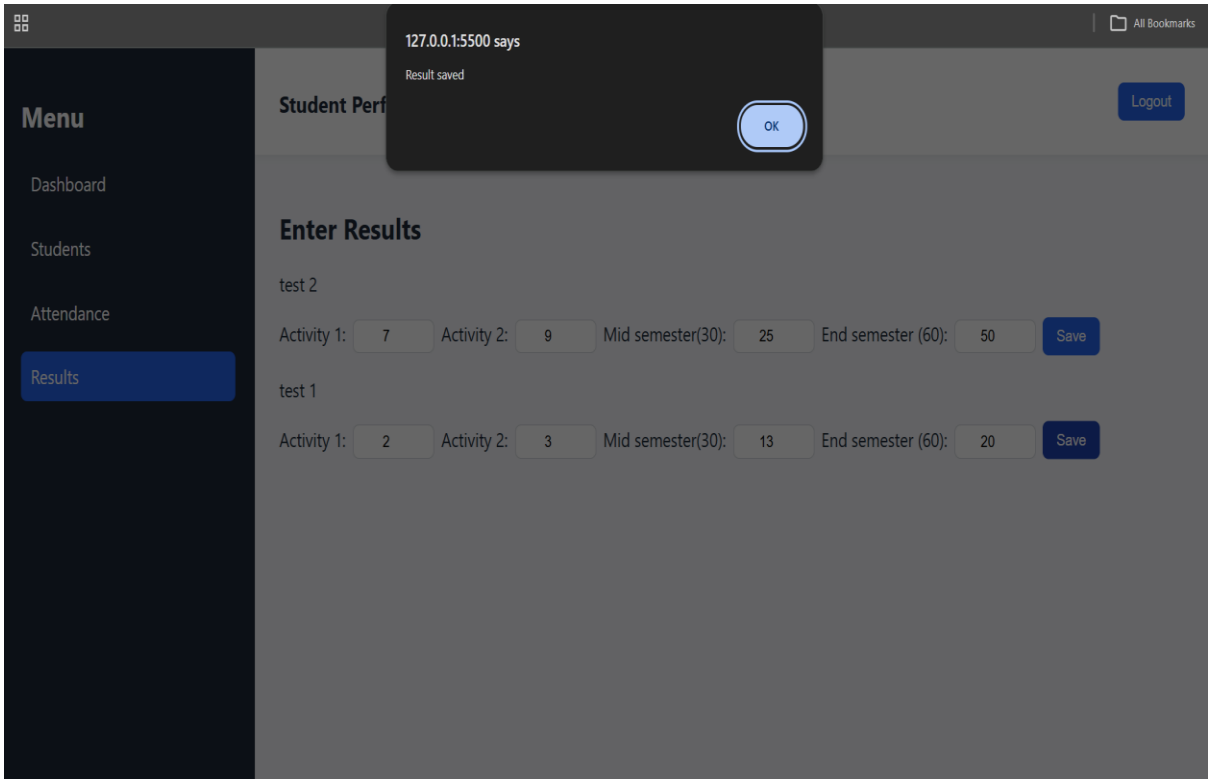


Results Page

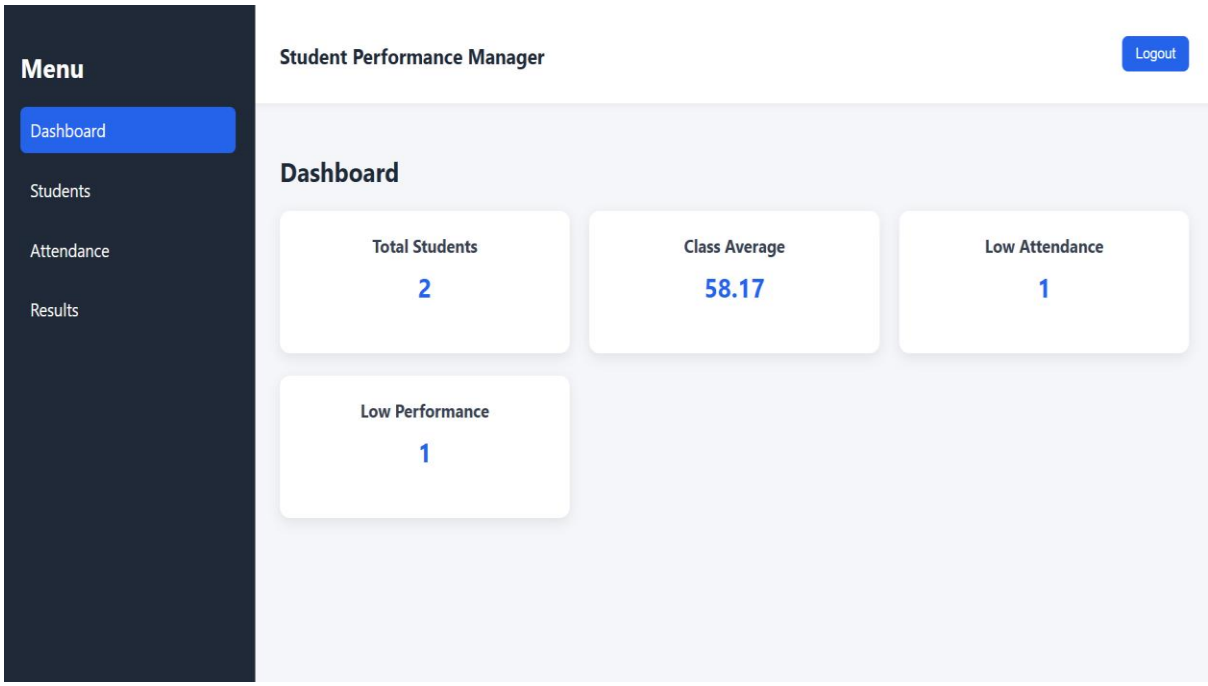
The results module allows teachers to enter internal assessment marks and exam marks. The system automatically calculates the final score.



Results **saved** to database

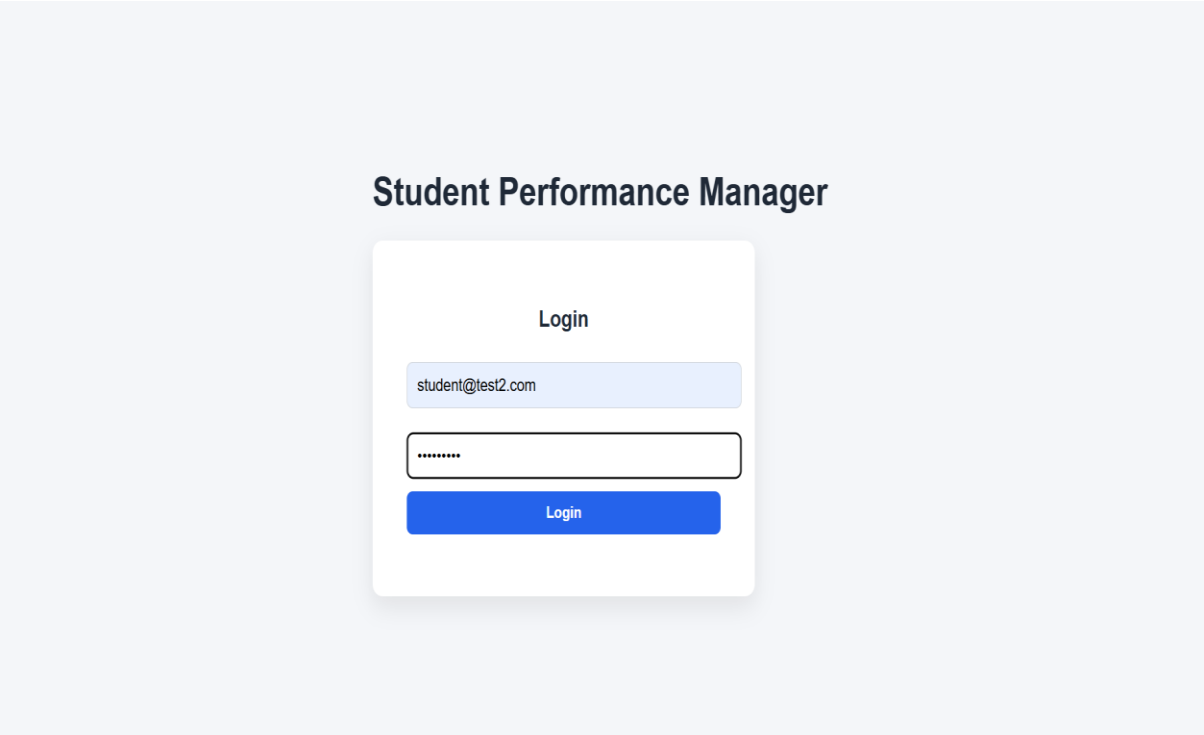


Final teacher dashboard after the above operation. Logout button will lead back to login page

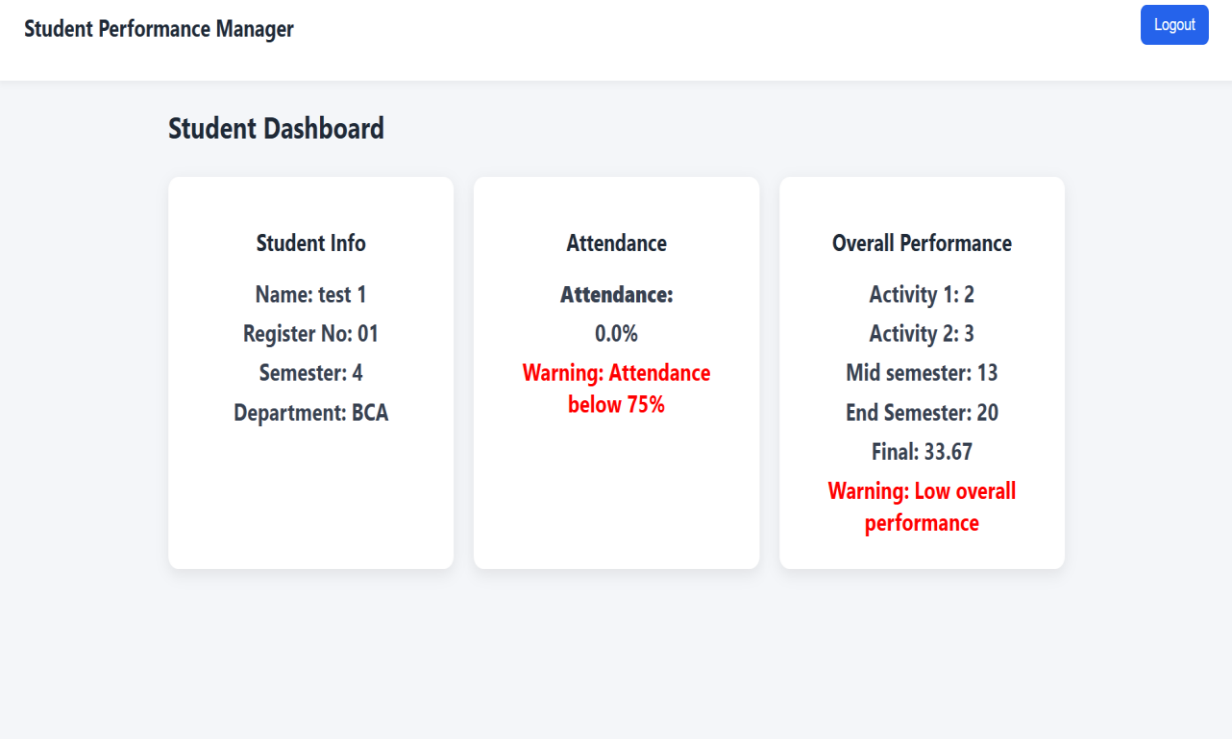


Student-side View:

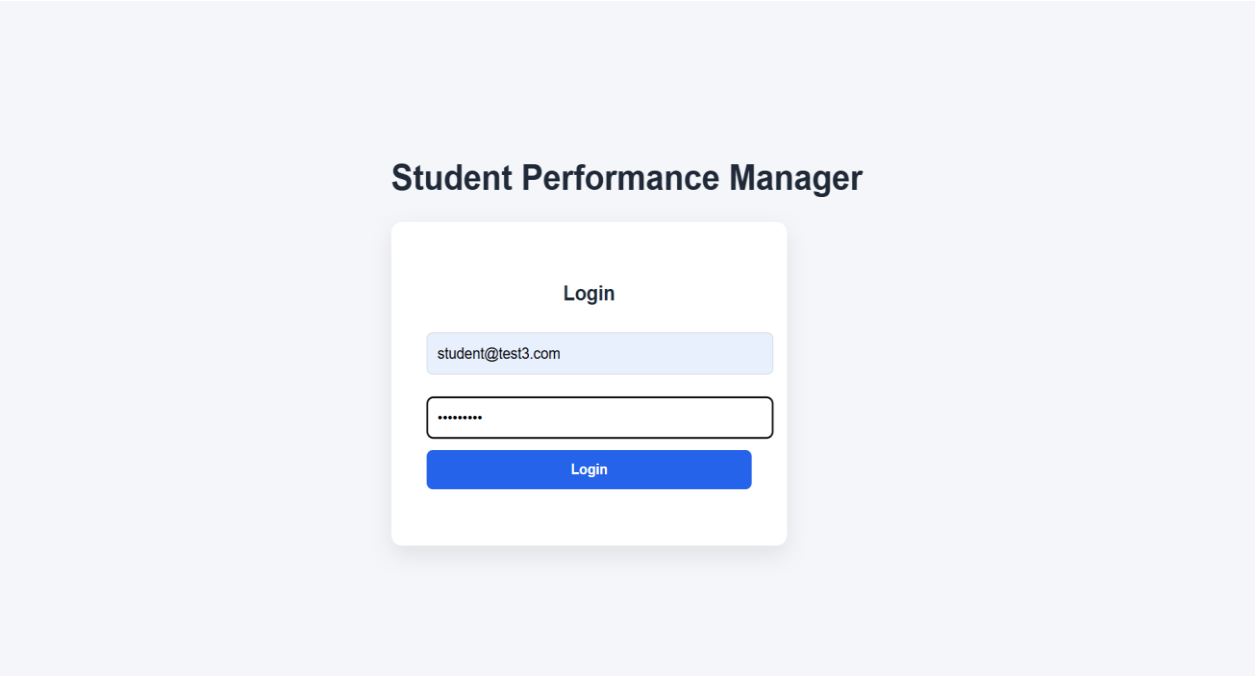
1. **Student 1** with attendance lower than 75% and results below 40 will get a **warning message.**

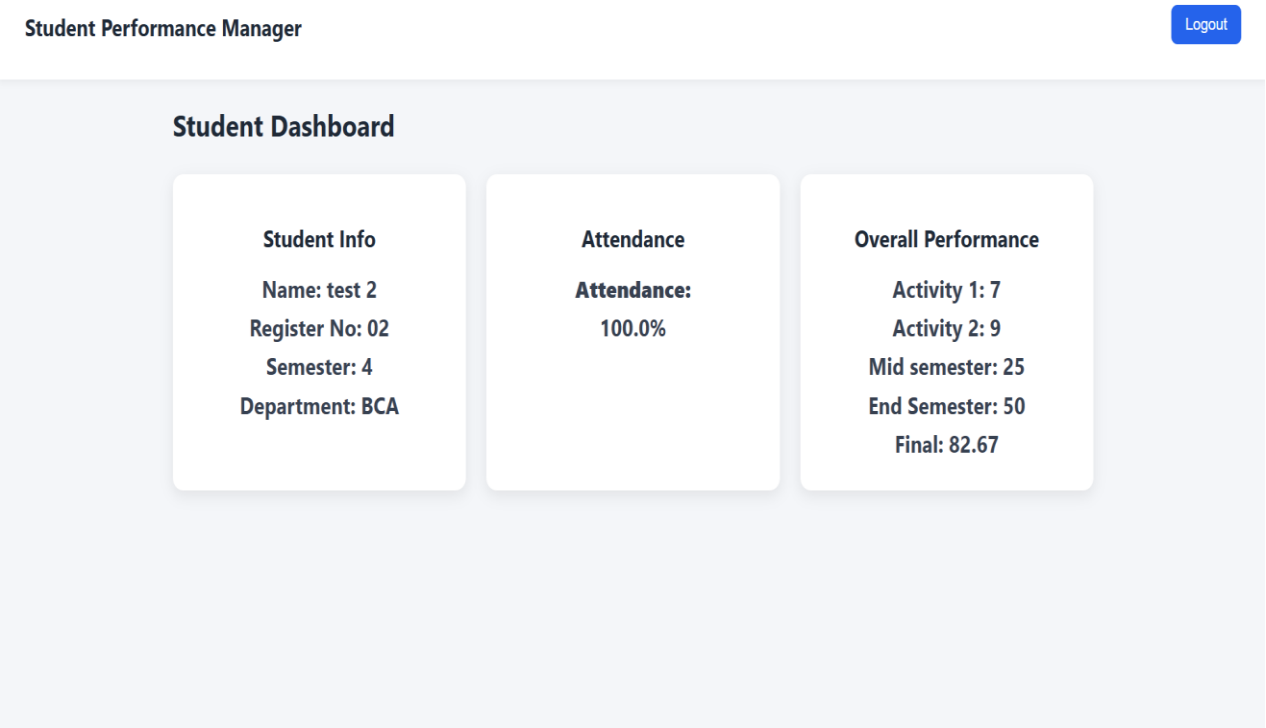


With warnings

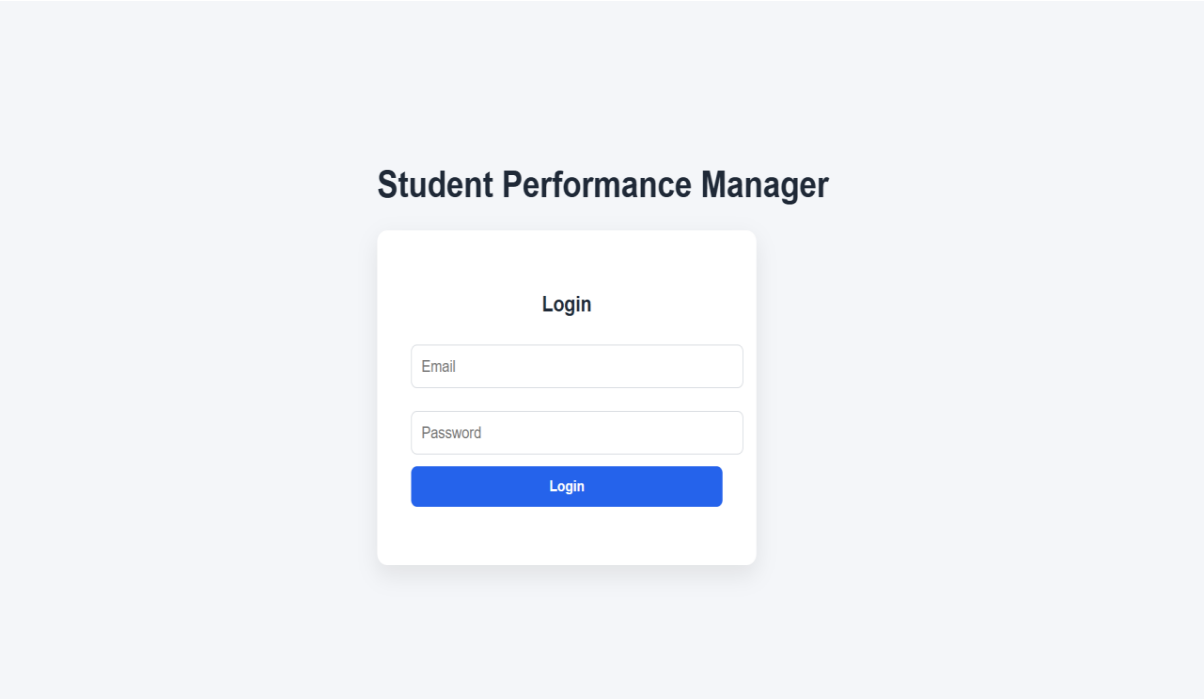


2. **Student 2** dashboard with no warnings


Student Performance Manager
Login
student@test3.com
Login

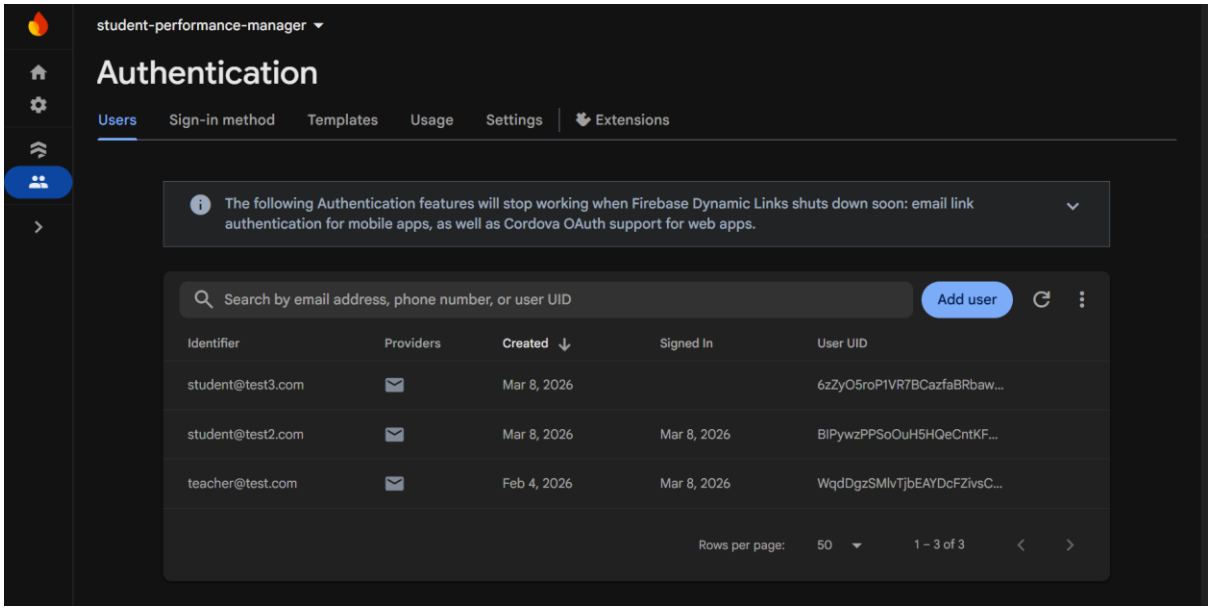

Student Performance Manager
Logout
Student Dashboard
Student Info
Name: test 2
Register No: 02
Semester: 4
Department: BCA
Attendance
Attendance:
100.0%
Overall Performance
Activity 1: 7
Activity 2: 9
Mid semester: 25
End Semester: 50
Final: 82.67

Logout button leading back to login page



Firebase Database:

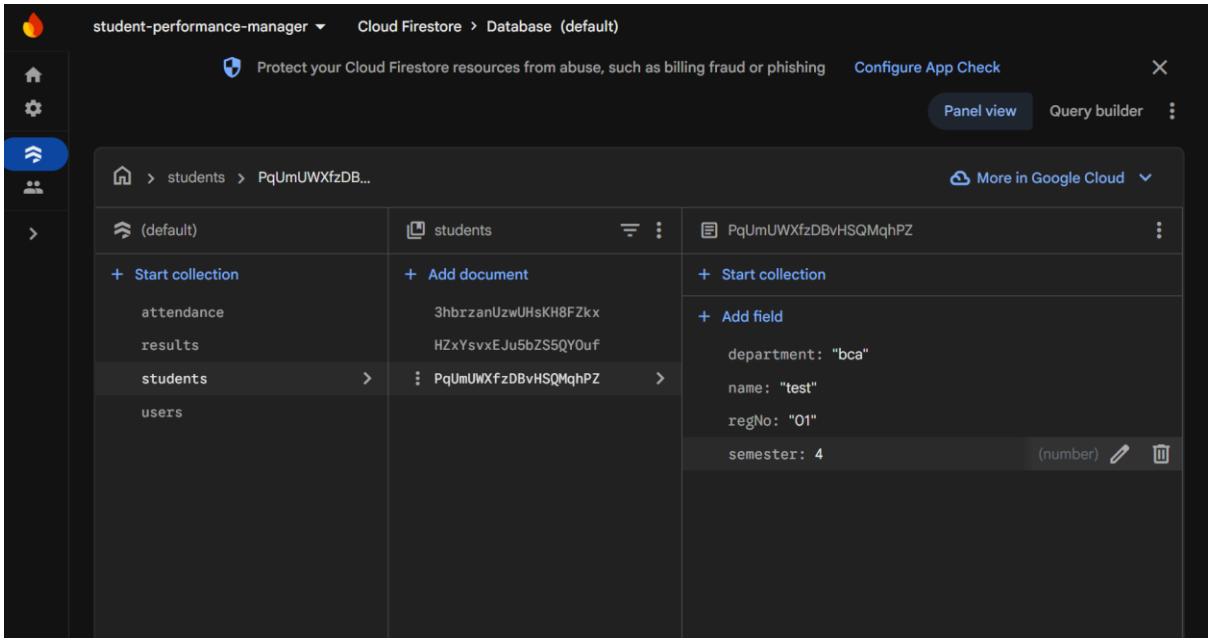
Users like teachers and Students created in the firebase database with email.id and password for login



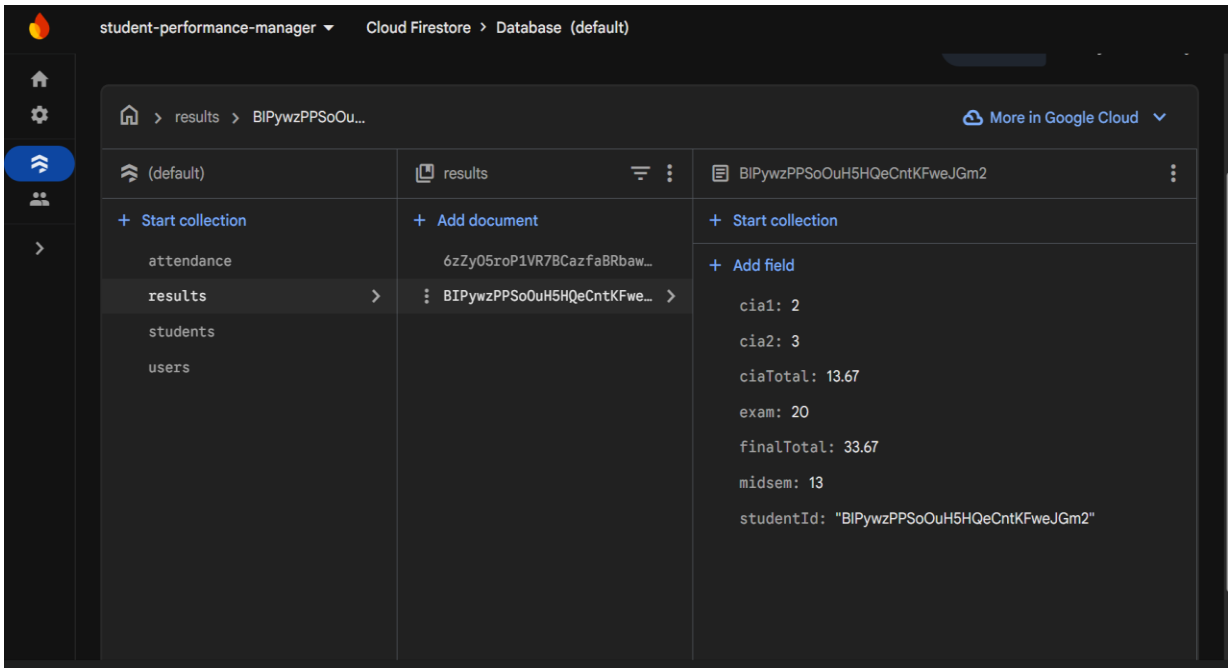
Firebase collections:

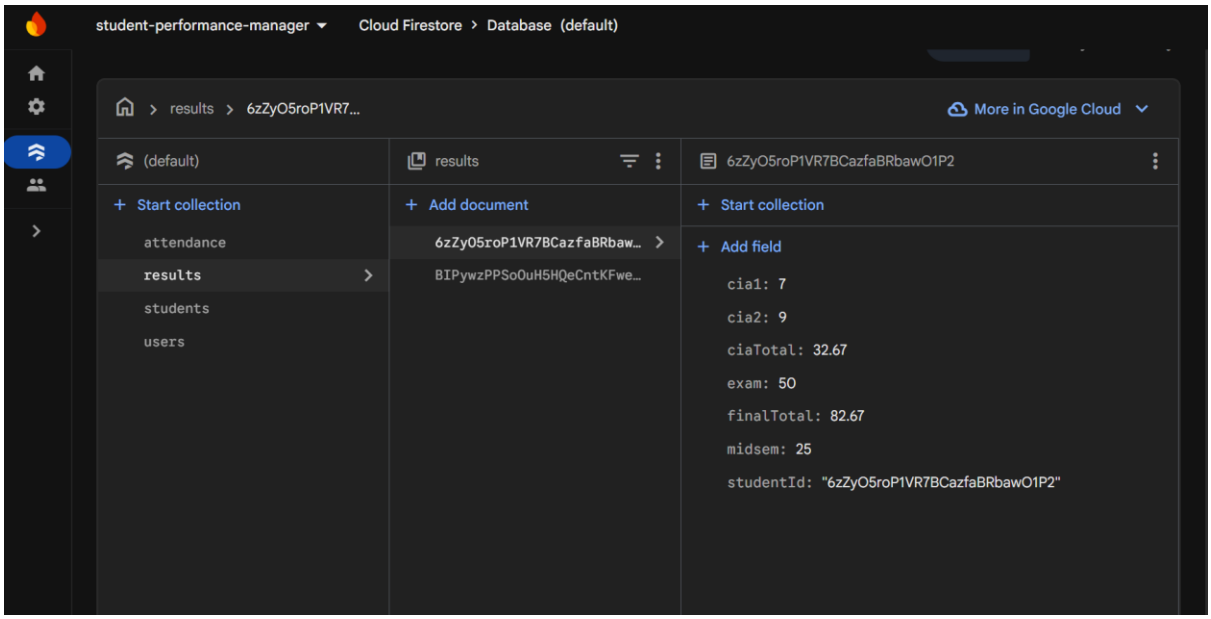
1. students collection for students page and CRUD operations.

Any student added in the website will be simultaneously saved in the database in **students collections** with unique IDs.



2. **Results collections**
   The activity and exam marks entered for each student in the website will be saved here in the Firestore.

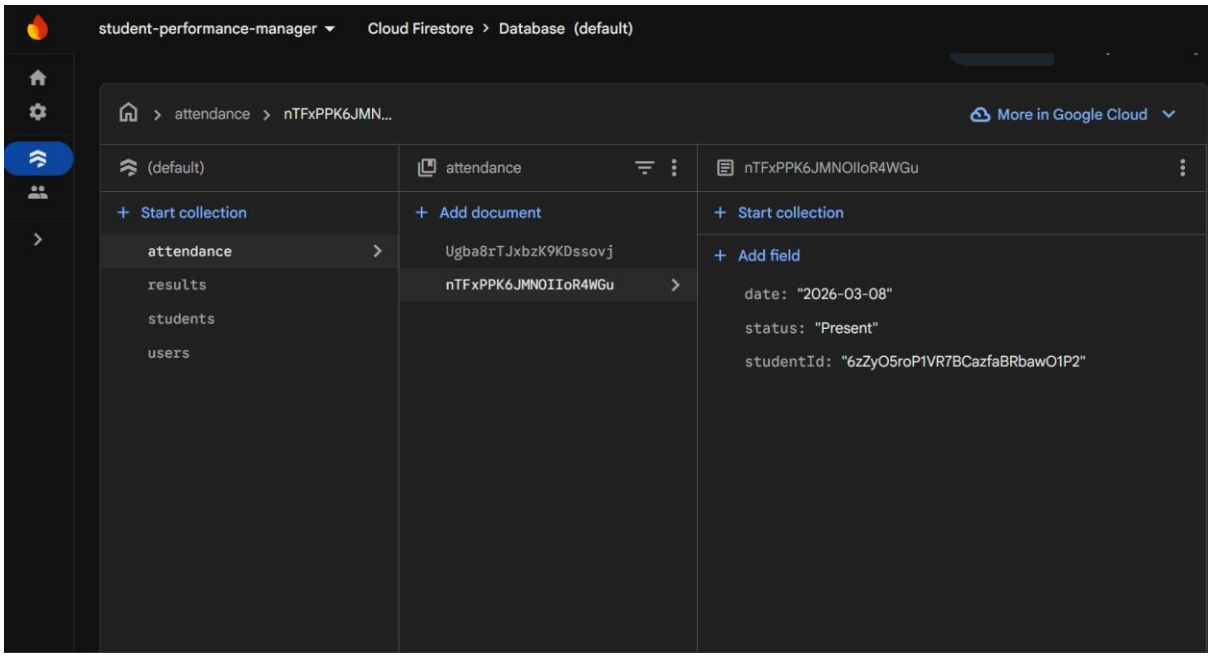




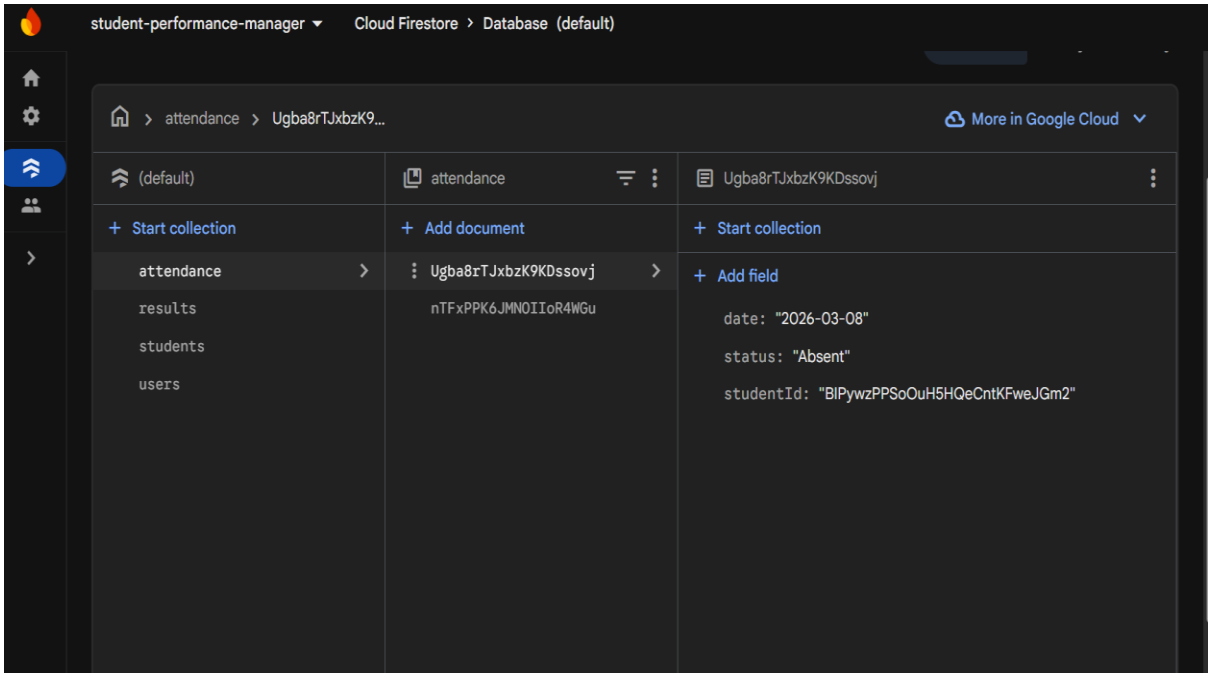
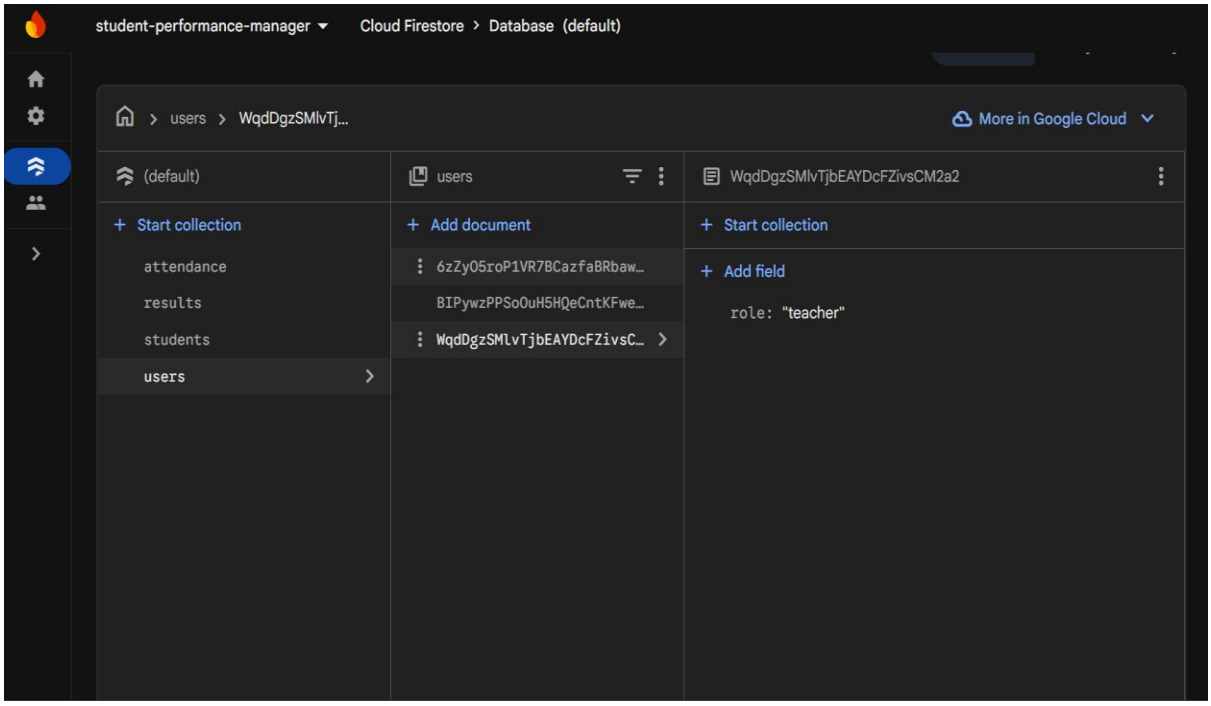
3. Attendance collections:

The database will automatically update the students' attendance in the database after each entry and save in the website. Each document in the database has an **auto ID** to save per day attendance for per student.

Test 2 student marked **present** saved in the database



Test 1 student marked **absent** saved in the databse



4. **Users Collection:**
   Each user created is added to users collection with their unique ID to help login with authorized access.